

# Florian Wiesner

[www.flowsnr.github.io](http://www.flowsnr.github.io)

## EDUCATION

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- **Dr.-Ing. (PhD) in Machine Learning & Chemical Engineering** May 2021 – Feb. 2026  
*RWTH Aachen University*  
*Summa cum laude*  
Aachen, Germany

## PUBLICATIONS

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- **Towards a Physics Foundation Model** Feb. 2026  
*Under revision at ICML 2026, [blog post](#), [arxiv](#)*
- **Data Mining for Enhanced PEM Electrolysis** Jan. 2026  
*Journal of The Electrochemical Society, [doi: 10.1149/1945-7111/ae335e](#)*
- **Modeling of Gas Diffusion Electrodes** Jun. 2024  
*ACS Applied Materials & Interfaces, [doi: 10.1021/acsami.4c04641](#)*
- **Electrowetting and Reactions in Microfluidic Gas Diffusion Electrode** Feb. 2024  
*Small, [doi: 10.1002/sml.202310427](#)*
- **Additive Manufacturing of Intertwined Electrode Pairs** Oct. 2022  
*Advanced Engineering Materials, [doi: 10.1002/adem.202200986](#)*
- **Process Model for High Salinity Deionization** Dec. 2020  
*Journal of Membrane Science, [doi: 10.1016/j.memsci.2020.118614](#)*

## EXPERIENCE

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- **University of Virginia** Feb. 2025 – Now  
*Visiting Scholar at the [Visual Intelligence Laboratory](#)* Charlottesville, VA, United States
  - **Multi-modal AI for Science:** Research on multi-modal datasets and models (physics & language)
  - **Physics Foundation Model:** Developed a Physics Foundation Model capable of modeling unseen physical systems purely from previous context.
  - **Research supervision:** Supervised research of two PhDs regarding Graph Neural Networks and Autoencoders.
  - **Student mentoring:** Mentored PhD students, fostering software engineering best practices and presentation skills.
- **RWTH Aachen University** May 2021 – Feb. 2026  
*PhD, Researcher* Aachen, Germany
  - **Group Leader TriggerINK Robotics:** Led the [TriggerINK](#) Robotics research team (5 PhDs) focused on robotic in-wound 3D-printing of an interactive bio-ink for supporting cartilage regeneration.
  - **Object Tracking with Synthetic Images:** Developed a mask-RCNN model trained on synthetic microscope images (rendered using Blender) to track microgels. Evaluated microgel flow using SORT and multi-hypothesis tracking algorithms.
  - **Thesis Supervision and Mentorship:** Managed 20+ undergraduate and graduate students and 5+ research assistants, fostering technical and personal development.
  - **Research Cooperation with University of Alberta:** Initiated a collaborative research effort between RWTH and the [ESD Lab](#) at the University of Alberta, led by Prof. Marc Secanell.
  - **Research Project Management:** Managed a research project which was part of the [H2Giga](#) flagship project. The project had an investment budget of 500k Euro, a yearly budget of 10k Euro and a duration of 4 years.
  - **Teaching and Lectures:** Delivered lectures and tutorials for 3 years, enhancing student engagement.
  - **Lab Management:** Managed a large research laboratory (wet-lab) with around 10 active daily users.

## PERSONAL PROJECTS - [GITHUB](#)

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- **ML-training suite:** Boilerplate pytorch code for DDP, compile, amp, checkpointing, time keeping on HPCs.
- **CUDA kernel:** Custom CUDA kernel for masked convolutions.
- **N-body system simulator:** Toy C++ simulator for planetary movements.